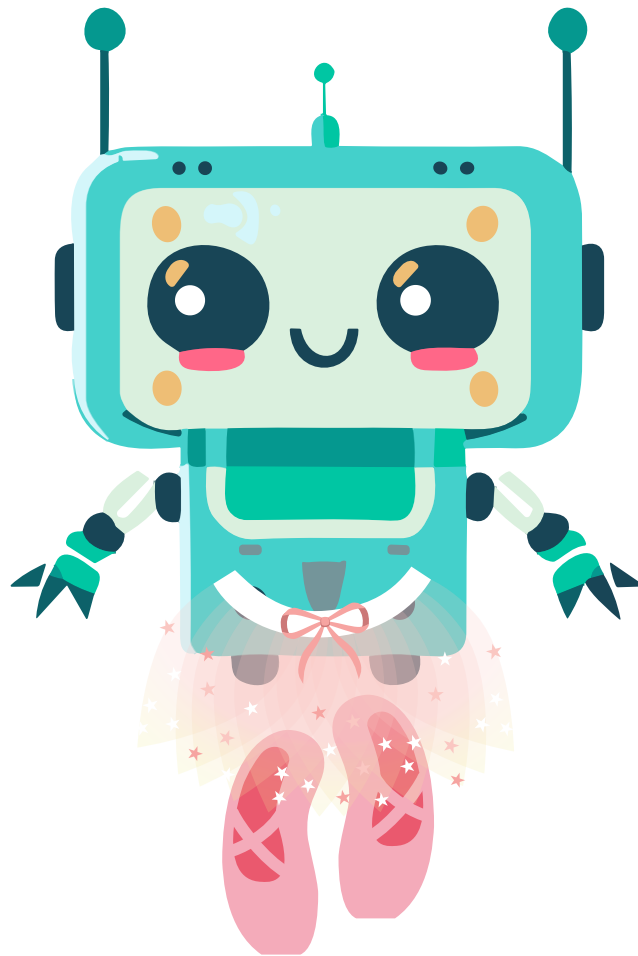
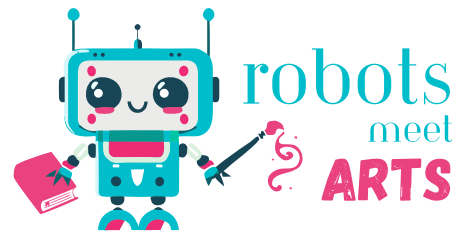




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Robot ballet



Subject: programming a robot
robot ballet

Engaging project for kids to learn Accurate Robot programming.

Computational practices:
choose what fits

- Decomposing
- Pattern recognition
- Abstraction
- Problem solving

Unplugged/**plugged:**



Preparation time:
10 min



Age:10- 12 years



Duration:
75 min



Location: classroom

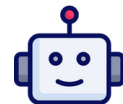


Group Distribution:

This relies on the number of accessible robots.



Difficulty (if already worked with robot)



Following this lesson, the children will have the ability to:

- name the characteristics of ballet (history, music, movement)
- Create a basic code for a robot
- Investigate reasons why a robot may not be functioning properly, identify the issue, and attempt to resolve it
- Experiment with different coding approaches and learn how to integrate the new knowledge effectively.

Materials Needed:

- LEGO WeDo 2.0 (any other robot can also be used: beebot, lego,...)
- Laptop with the LEGO WeDo 2.0 software installed

Helpful links:

General website LEGO WeDo 2.0: <https://education.lego.com/nl-nl/>
software :

<https://education.lego.com/nl-nl/downloads/retiredproducts/wedo-2/software/>

video ballet: <https://www.youtube.com/watch?v=0EekFTj5PvU>

Preparation and teacher notes

Make sure the software is already installed on the laptops.

Warm up + build up 30'

Today, we will discover ballet. We will try to let a robot dance some ballet. 😊 If you already know a lot about ballet, please share with the rest of the class!

This video (see page 2 for helpful links) will provide us with a brief **history of ballet**.

Do we already have some insights into the characteristics of ballet? Write down the answers in a **mindmap**.

Quote an important part of the video (4.03):
'Fortunately, today most of us don't have to learn a complicated set of steps just to socialize at a wedding. Instead we can go to a-the theater to see professionals who spend their lives training rigorously.'

Why are we going to use robots to create a ballet dance?

→ robots are 'easy' / fast to train.

Now we focus on **how to dance ballet**. Get on your feet and show me your best impression of a ballet dance. Annex 1 can help.

What are the characteristics of ballet dancing? Complete your **mindmap**.

Finally, we study the **ballet music** Annex 2 can help. Complete your **mindmap**.

Time to enjoy the music!

- make gouns (+- 3, This relies on the number of accessible robots.)
- **discover songs** by one of three well-known composers:
 - Pyotr Tchaikovsky
 - Igor Stravinsky
 - Ludwig Minkus
 -
- Let them choose their favorite song after 5 minutes.

Preparation and teacher notes

Text

Rehearsal 30'

Today we make robots dance ballet. We have just chosen our perfect ballet music. Take a look at our mindmap. What features can a robot mimic to dance ballet? Color green.

- synchronized dancing
- memorize the choreo
- turns fast/slow
- ...

Let's get started!

- open Lego WeDo 2.0 on the computer
- build an easy code and syncornize with the robot
- Yay! My robot moves 🤖

Get creative!

- select 10 seconds of the song
- write code that fits the music → **Try different codes until you find the prefect dance** 🧠

Act like a pro!

- Rewrite the code on another laptop and sycronize with another robot.
-
- **Try to get both robots to dance in sync.**
 - Why is this so hard?
 - they must start at exactly the same time
 - The code should be exactly the same
 - Don't give up keep practicing just like real ballerinas

Final show + clean up 15'

Watch your classmates' choreos, give them well-deserved applause! ✨

Annex 1

✨ Graceful Movements

👣 Pointed Toes

🎭 Expressive Storytelling – Ballet often tells a story using movements and facial expressions instead of words.

👤 Strong Posture – Dancers stand tall with straight backs and open arms to look poised and confident.

🔄 Turns and Jumps – Ballet includes amazing spins (pirouettes) and high jumps (jetés) that take lots of practice!

Annex 2

Ballet isn't just about dancing—it also has wonderful **music that helps tell magical stories**. Let's look at some special things about ballet music!

1. Ballet Music Tells a Story! 📖

- Ballet music is made to **help dancers tell a fairy tale**.
- The music changes to **match the feelings** and actions of the dancers.

2. It Started a Long Time Ago! ⏳

- Ballet music began in the **1500s in France and Italy**.

3. The Orchestra Brings It to Life! 🎻

- Ballet music is played by a **big group of musicians** called an **orchestra**.
- Instruments like **violins, flutes**, and drums make the music exciting and **magical**.

4. Famous Composers Made Beautiful Ballet Music! 🎵

- Some very talented musicians wrote ballet music, like:
 - Pyotr Tchaikovsky – He wrote The Nutcracker and **Swan Lake!** 🦢
 - Igor Stravinsky – His music made ballet even more powerful and exciting!
 - Ludwig Minkus – He wrote Don Quixote and La Bayadère!

5. Different Styles for Different Stories! 🧠

- Some ballet music is **soft and dreamy** (like Sleeping Beauty 🛌).
- Some is **fast and dramatic** (like The Firebird 🔥).

6. Music Helps Dancers Move! 🩰

- Ballet dancers listen closely to the music to know **when to jump, spin, or glide**.

7. Ballet Music Is Still Popular Today! 🎧

- Even today, people love ballet music in movies, concerts, and **theaters!**
- You can hear famous ballet songs in cartoons, commercials, and even video games!

Isn't ballet music amazing? 🎵 Now, let's listen to a beautiful ballet song and pretend to dance like real ballerinas! 🩰✨